

Fairweather · Workflow note

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Live: <https://fairweather-plum.vercel.app>

Repo: <https://github.com/debugagu/fairweather>

The note

What the app decides, where the numbers come from, and the choices behind them.

WHO THIS IS FOR

Someone walking outdoors for most of the day, away from shelter. A day hiker.

Every threshold below follows from that. Someone on a hike cannot step inside when the sun turns, cannot leave a ridge quickly, and is working hard the whole time. Thirty-two degrees is a pleasant afternoon if you are moving between air-conditioned shops and a real problem four hours into a walk. Numbers tuned to cover both cases would be too vague to act on, so I picked one.

THE WORKFLOW, END TO END

Everything below runs in the browser on each search, and nothing is kept between them.

1. City name to coordinates

The typed name goes to Open-Meteo's geocoding endpoint asking for up to five matches.

- **No matches.** An error state suggesting a country be added, since that is usually what fixes it.
- **One match.** Used, and the resolved place is printed above the results so you can see which one it picked.
- **Two or more.** A separate screen listing each option with its region, country, population and coordinates. "Springfield, Illinois, population 114,394" is enough to choose between; "Springfield" is not.

Population is shown because it is the quickest way to recognise the right place. It is not used to rank or auto-select. Picking the first result automatically would be less code, but it guesses on the user's behalf.

2. Coordinates and dates to a forecast

Sent with `timezone=auto`, so days break at local midnight rather than the reader's. Twelve daily fields come back, plus two hourly series.

Dates are bounded at both ends: no earlier than today, no later than fourteen days out. The same rules are checked again in JavaScript before any request is made, because a native date picker can be bypassed. Past dates are rejected rather than redirected to the archive API, since a forecast and a historical record are different things.

3. Height, if you gave one

A forecast describes a point on the map, and that point is the town. A hiker starts there and climbs. Chamonix sits at 1,060m; the paths above it run past 2,400m. Reading the valley forecast and walking the ridge is how people end up cold, so if you say how high you are going, every figure is re-derived for that height before any threshold sees it.

4. Seven concerns, scored separately

Rain, heat, cold, wind, UV, daylight and the snow line. Each returns a level: 0 not worth mentioning, 1 worth knowing, 2 changes the plan, 3 do not go. The day takes the highest level reached.

5. One concern writes the answer

The highest-scoring one. Ties break on a fixed order (heat, wind, cold, rain, UV, daylight), ordered by what is most likely to cause actual harm.

Each day shows three tiers, in this order and this size order. The call (**Good day, Mostly fine, Plan around it, Do not go**) is the largest thing on the card. The reason sits in one line under it. The numbers are the smallest thing there. If you have to read a temperature to know what the app is telling you, the hierarchy is wrong.

6. The rest go to the packing list

Concerns that did not write the sentence still contribute items. A day whose verdict is about heat will still put a waterproof in the bag if there was rain in it.

THE THRESHOLDS

SIGNAL	LEVEL 1	LEVEL 2	LEVEL 3	REASONING
Rain	≥35% or ≥1mm	≥60% and ≥2mm, or ≥5mm	≥70% and ≥10mm	Both signals are required, because probability on its own is misleading. A 70% chance of 0.5mm is nothing; 12mm at 55% soaks through anything that is not a proper shell.
Heat	28°	32°	38°	Apparent temperature rather than the thermometer, since it accounts for humidity and that is what decides how hard the day feels. Above 38° this stops being about comfort.
Cold	7°	2°	-5°	Apparent minimum, which is what you meet at dawn and at rest stops. While moving you generate your own heat. 2° means ice on the path.
Wind	25 km/h	40 km/h	60 km/h	Gusts rather than the sustained average. The average affects how tiring the day is; the gust decides whether a route is sensible.
UV	6	8	11	Exposure here lasts all day with no shade to step into. Above 8, sunscreen alone does not survive several hours of sweat, so the advice changes to covering up.
Daylight	<9.5 h	n/a	n/a	Capped at level 1. Short days never make weather bad, they just limit how far you can go.

ALTITUDE

This is the part I would defend hardest, because it is the difference between a forecast and advice for the person using it. Every weather tool gives you the valley. A hiker spends the day well above it.

Give the app a height and it recalculates before judging, so the existing thresholds end up asking about the right place. Chamonix on a settled September week reads "A good window for it. All 4 days are walkable with nothing to plan around." The same city and dates at 2,800m: "A demanding week. Only 1 of the 4 days is straightforward." Hard frost at -4°, and gloves appear in the packing list. Nothing about the judgement changed, only the question.

What is adjusted, and on what authority

- **Temperature**, at the environmental lapse rate of 6.5°C per 1,000m. That is the standard atmosphere figure. Real rates run from about 5 in saturated air to 9.8 in dry, so it is a middle estimate rather than a measurement, and it is applied to the feels-like figures too.
- **UV**, at 10% per 1,000m. Less atmosphere overhead means more gets through, and published estimates cluster between 8 and 12% per kilometre.

What is not adjusted, and why not

- **Wind**. A ridge is far windier than the valley, but by how much depends on the shape of the ground, and no honest figure comes out of a single elevation number. A made-up multiplier would look more precise and be less true, so the app says on screen that the wind reading is a valley one.
- **Rain**. More falls with height, but the part that matters is whether it lands as rain or snow, and the freezing level answers that directly.

The snow line is fetched, not estimated

Open-Meteo returns `freezing_level_height` hourly, so this is the one altitude judgement resting on a measurement. Averaged across daylight and compared against your target height, it becomes the seventh concern: 400m or more above it means frozen ground and traction in the pack; within 100m means you cross the line partway up and should know where. It is deliberately separate from the cold concern, which is about air temperature. This one is about what is under your boots.

HOW THE WRITING WORKS

Every verdict is three sentences, in the same order:

1. What kind of day this is.
2. What the number actually means once you are out in it.
3. What to do about it.

The middle sentence is the one that earns the product. A forecast can already tell you 6mm. It cannot tell you that 6mm soaks a fleece through and stops there, which is the part that decides whether you go.

Plan around it. You will get wet, but it is still a day out. Around 6mm at 70%, which is enough to soak a fleece through and not much more than that. Shell on from the start, and keep one dry layer in the pack.

Good days were the harder ones to write. Most days on most trips are unremarkable, so a single sentence repeated down the page reads as filler however well written it is. The good-day text branches on conditions that are real but never bad enough to become a concern: a bit of breeze, moderate sun, a wide or narrow temperature swing, a long or short day. Across a settled eight-day spell that produces eight different paragraphs. The closing sentence is dropped when it has nothing to add, so some days run to two sentences and some to three, because uniform length is its own kind of tell.

SEVERAL PROBLEMS ON ONE DAY

The advice has to stay short. The highest concern writes it, and at most one other gets a three-word fragment after it, only if it is serious enough to change what you do, or if the day is mild enough that two small notes fit:

"Front-load this one. It climbs to 34° by early afternoon, and on open ground there is nowhere to sit it out. Start at first light and be somewhere cool by midday. The wind will fight you too."

Three are never listed, because past two it turns back into a weather report.

UNREMARKABLE DAYS

Most days are unremarkable, so they get the quietest card on the page and text that branches on the small differences between one settled day and the next:

"Cold but clean. 8° all day and dry, which is better walking weather than it sounds. Barely a few degrees in it from dawn to dusk, which makes packing simple."

Not flagging a day is itself useful information. It is also why colour only appears at levels 2 and 3; marking every day would make the marks useless.

THE ICONS



Seven glyphs drawn as monochrome strokes, mapped from the WMO weather code Open-Meteo returns and collapsed from about thirty codes, since a 21px glyph cannot show the difference between light and moderate drizzle.

They are descriptive only and feed into no threshold. An overcast day still reads "good walking weather", because cloud has little to do with whether a walk is worth taking.

THE FIVE STATES

STATE	WHEN	WHAT IT SHOWS
Empty	Before any search	What the tool does, what you get, and what to type
Loading	While a request is in flight	A skeleton, plus which city is being looked up
Choosing	Two or more places share a name	Each candidate with region, country, population, coordinates
Nothing found	The city does not exist	A plain notice, deliberately not styled as a failure
Error	Bad dates, API down, rate limited	What went wrong in words, and a retry
Results	A forecast came back	The report

Nothing found and Error are separated on purpose. A search that returns nothing is not a crash, and colouring it red makes people think the app broke.

THE NUMBER I WOULD WATCH

Not visits, and not time on page, because both go up when a product is confusing. The number that would tell me this worked is the share of searches where someone changes their dates and searches again. Reading a forecast is not the point. Changing a plan is. That is the only behaviour showing the verdict actually did something. Second to that: whether anyone comes back for a second trip.

COLOUR

The page sits on a blue gradient, and exactly one colour carries meaning: the rust used on days scoring 2 or 3, on their marker, their glyph and their heading. It appears nowhere else. The blue is a background rather than an accent, so the rule of one working accent still holds. Good days get no colour at all, because marking every day would make the marks useless.

The hourly chart. The app fetches an hourly series and draws none of it. Most weather tools put a 24-hour graph on screen and leave you to interpret it. Here the same data produces one instruction instead: "The rain clears around 13:00, so there is a good afternoon in this if you start late."

Three smaller omissions. Cloud cover is drawn as a glyph but never scored. Humidity is not shown separately, because apparent temperature already accounts for it. And the packing list covers weather only, with no boots, map or first aid; listing things you would bring anyway makes the list longer and less useful.

WHAT I WOULD BUILD NEXT

1. **Confidence.** A ten-day forecast is currently treated the same as a two-day one. Comparing model runs would let the wording soften as the horizon extends.
2. **Wind that knows about terrain.** Altitude adjusts temperature and UV, but wind is still a valley reading, because a single elevation figure cannot tell you how exposed a ridge is. Terrain data could, and wind is the signal most likely to turn a hard day into a dangerous one.
3. **A second audience.** Not a toggle over these numbers, but a separate set of thresholds. Trying to advise a hiker and a wedding guest from one table would make both worse.

BUILT WITH

Plain HTML, CSS and JavaScript. Six ES modules and nothing to install. Geist and Geist Mono, with figures set in the mono so the columns hold still between days. Deployed as static files on Vercel, which needs no configuration because there is nothing running on the server side.

Data comes from Open-Meteo's free geocoding and forecast endpoints, called live from the browser on each search. Both are open, so there is no key to leak.

Built with AI assistance, using Claude Code. It was quickest at the parts that were already decided: markup, CSS, and the plumbing around two API calls. The rest took longer. Who this advises, where every threshold sits, what to leave out and how a verdict should read are my calls, and a fair share of the work was cutting text that read as machine-written. The thresholds all sit in one file, so they are easy to argue with.